CSMFAB000000000106 Aegis Pipeline Redesign for Telluric Events

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to document:** `CSMFAB000000000106 Aegis Pipeline Redesign for Telluric Events V2.0`

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications invalidate V1.0 material property values
2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) vs ~115 (forecast used in V1.0); design basis updated accordingly
3. **Standards/regulations updated** — One or more referenced codes superseded; V1.0 compliance citations rendered obsolete

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V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Solar Cycle 25 risk framing	V1.0: Risk basis SSN ~115 (original SC25 forecast)	V2.0: Revised to SSN ~137 (NOAA SWPC Q1 2026 actuals); 10-year Carrington-class probability +20%	Observed solar activity exceeded forecast; design basis was non-conservative	NOAA SWPC Solar Cycle 25 Progression Report Q1 2026
GFRP pipe approval	V1.0: GFRP referenced; gas approval status pending	V2.0: ASTM D2996-2024 approves GFRP to 1500 psi; Aegis wall design to 2000 psi via Barlow	Approval gap resolved; specific pressure rating derivation added	ASTM D2996-2024; ASME B31.12-2025

Impact If V1.0 Retained (Criticality Matrix)

Parameter Changed	Consequence of Non-Upgrade	Severity
Solar Cycle 25 risk framing	Probability underestimated; design basis not conservative for observed SC25 activity	MEDIUM
GFRP pipe approval	Pressure certification gap; design could not be approved under V1.0 referenced standard	HIGH

Unchanged Elements

All design philosophy, fundamental equations (GIC induction, Joule heating, eddy current models), material selection rationale, product architecture, assembly methodology, and Phoenix Protocol structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMFAB000000000106 Aegis Pipeline Redesign for Telluric Events V1.0 archived; V2.0 is the active specification as of June 2026.